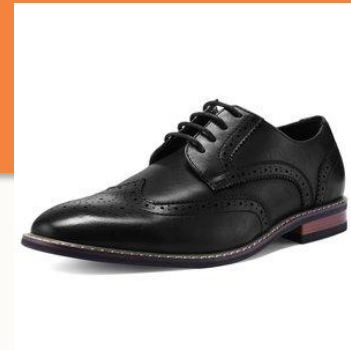




Dynamic Demand Modeling Using Multimodal AI

Price Elasticity Estimation for Amazon Shoes
via Deep Embeddings and Double Machine Learning



Jyoti Karna | Billy Saputra | Yin Yuhui | Ladityarsa Ilyankusuma

Advisor: Guy Arie, PhD | Ruidong Ma, PhD

Replication of Victor et al. (2025) "Adventures in Demand Analysis Using AI"

Northeastern University | April 2026

Agenda

01 Data Collection from Keepa

02 EDA, Cleaning & Train/Val Split

03 Multimodal Embedding Training

04 PCA, Clustering & Feature Engineering

05 DoubleML: Estimating Price Elasticity

06 Results: Women's vs Men's Comparison

07 Next Steps & Future Directions

Data Collection

Keepa API | Amazon US | Shoes Size 8 | Women & Men

keepa

Third-party Amazon price & sales tracker
Historical price, sales rank, Buy Box, ratings, reviews
Amazon does not expose this data directly
7 batch notebooks, split by sales rank range (1K-100K)
Date window: Jan 2025 to Mar 2026 (454 days)
Extended to Mar 2026 to avoid backfilling artifacts

amazon

Total ASINs	7,761
Total Rows	1,008,930
Date Range	Jan 2025 - Mar 2026
Women	5,472 ASINs
Men	2,279 ASINs

Both Women and Men analyzed independently using the exact same pipeline and methodology



EDA & Data Wrangling

Cleaning | Forward Fill | Stratified 50/50 Split

Women's Shoes

5,472 ASINs after gender filter

12 dirty ASINs removed

324 incomplete ASINs dropped post-fill

Final: 5,147 clean ASINs

Train: 2,573 | Val: 2,574

Top categories: Pumps, Flats, Heeled Sandals,
Loafers, Mules & Clogs, Ballet & Dance



Men's Shoes

2,279 ASINs after gender filter

11 dirty ASINs removed

217 incomplete ASINs dropped post-fill

Final: 2,061 clean ASINs

Train: 1,029 | Val: 1,032

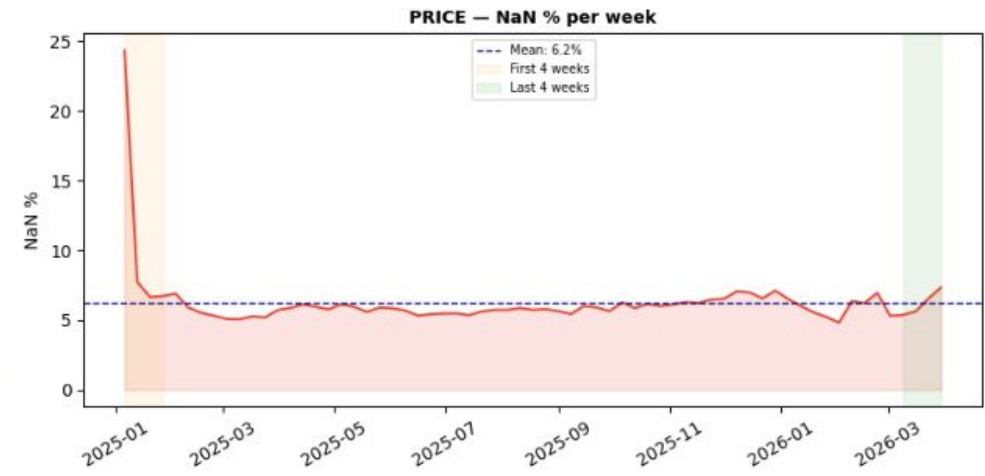
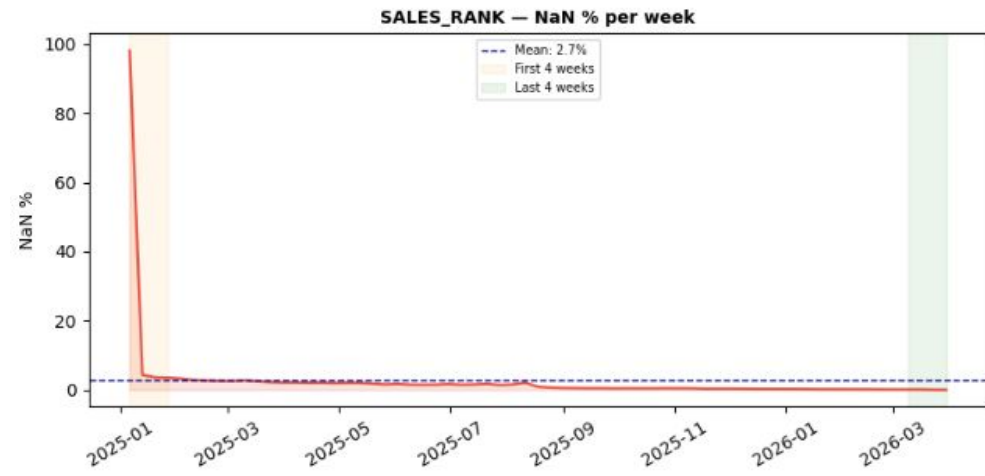
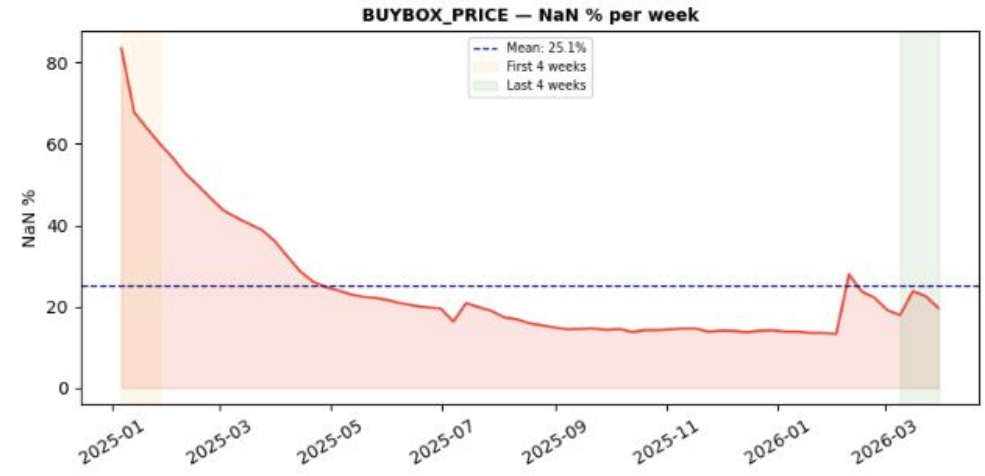
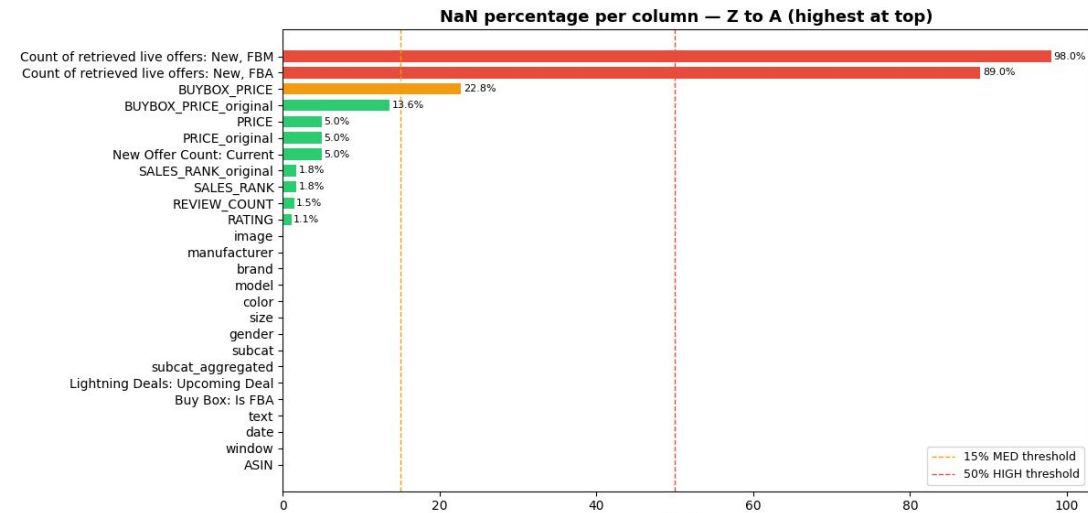
Top categories: Loafers, Oxfords, Sneakers,
Walking, Road Running, Trail Running



Raw data > Remove dirty ASINs > Gender filter > Forward fill (Jan/Feb seed) > Drop incomplete > Stratified split

EDA & Data Wrangling

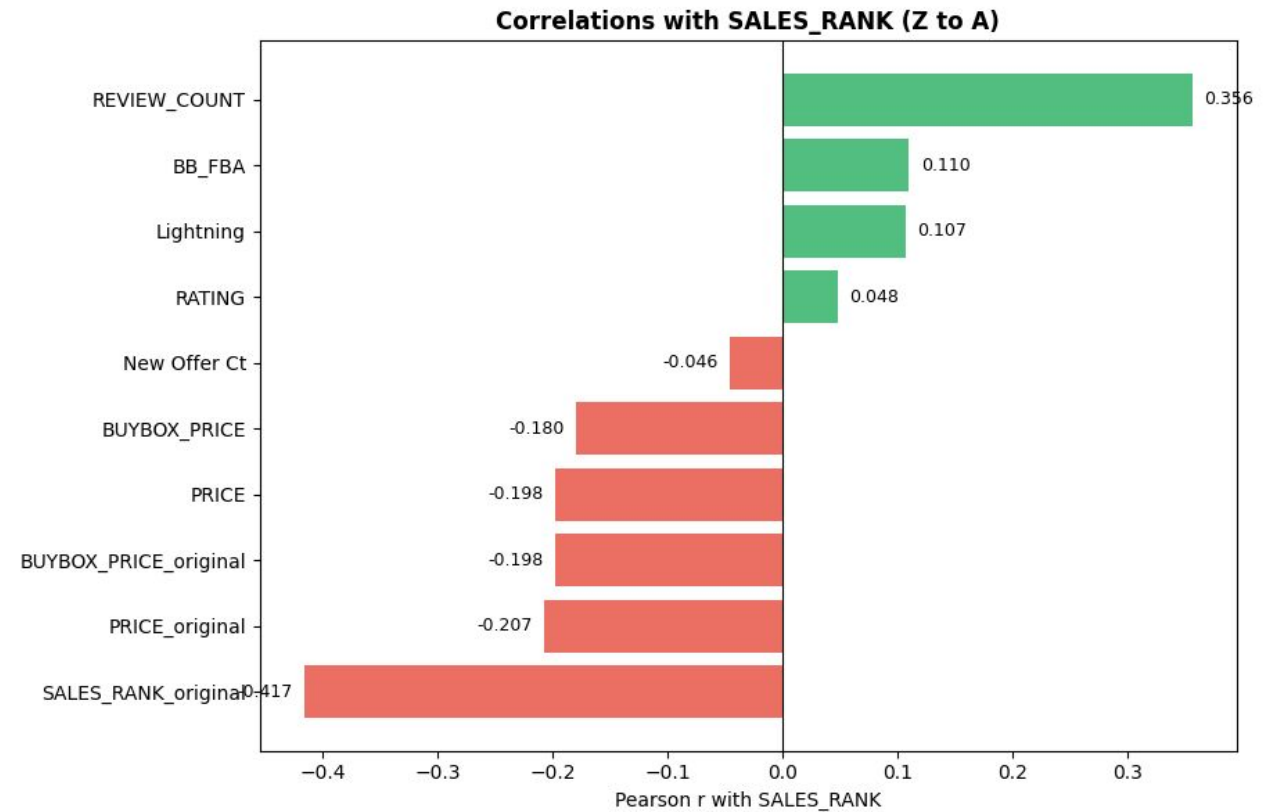
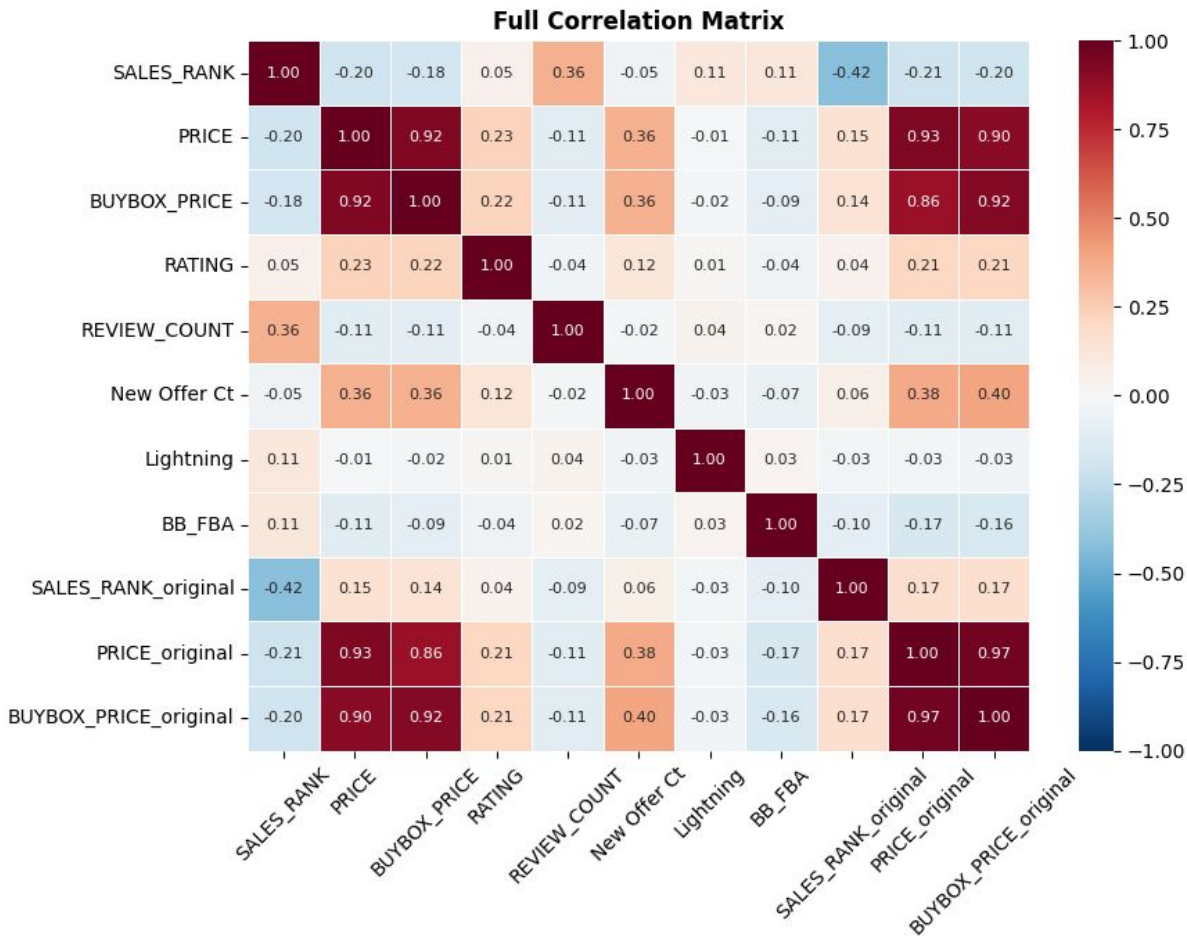
NaN Handling



NaN Value Distribution -> Removed January - February 2025

EDA & Data Wrangling

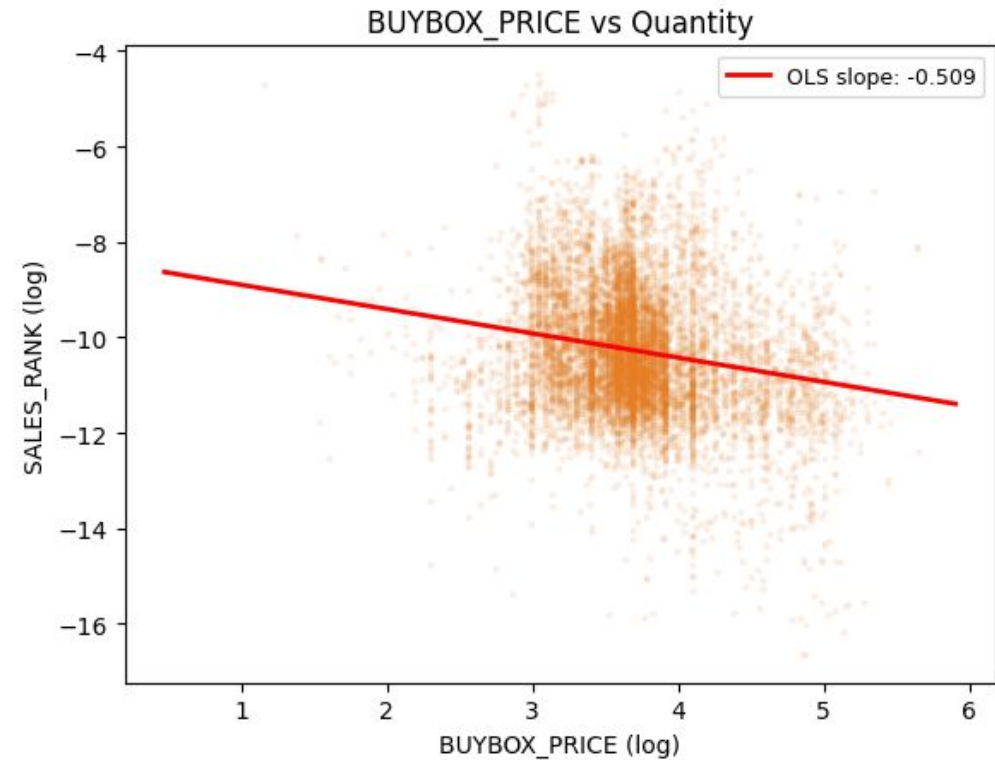
Feature Correlation



Weak correlation -> Need future selection for next step

EDA & Data Wrangling

Price vs Quantity Correlation



Price vs Quantity Relationship (Core for Demand Estimation)

Ordinary Least Squares (OLS) —> fits the best straight line to show the raw price-demand relationship

Slope -0.602 (PRICE) | Slope -0.509 (BUYBOX_PRICE)

However, OLS captures correlation, not causation, so we use DoubleML later to isolate the true causal effect

Multimodal Embedding Training

Victor et al. (2025) Appendix G | Three Encoders + Cross-Attention Fusion

Text Encoder

RoBERTa

Product title, brand,
model, color, size

Image Encoder

BEiT

Product photo
(224x224 pixels)

Tabular Encoder

SAINT

Rating, reviews, offers,
Buy Box, deals

Cross-Attention Fusion > LayerNorm > FC Projection > 128 / 256-dim Embedding

Sales Rank Prediction (Q)

Price Prediction (P)

4 Training Variants (NVIDIA H200 GPUs on Northeastern RC Cluster)

Part 3a

Text + Image + Tab
Lag-1

Part 3b

Text + Image + Tab
Time-indep.

Part 3c

Text only
Lag-1

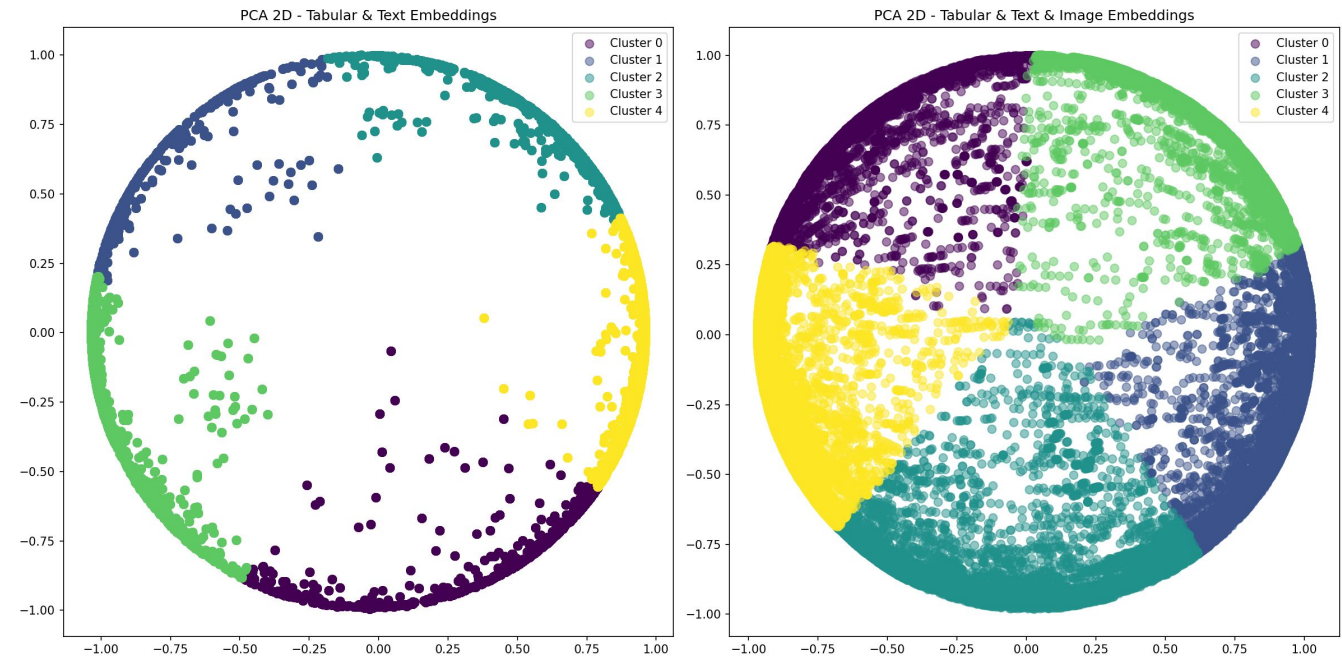
Part 3d

Text only
Time-indep.

Each variant produces 6 prediction zip files (2 embedding dims x 2 splits + 1 diff model x 2 splits) = 24 total

PCA & Product Clustering

KMeans (k=5) | Embedding Space | Women's Shoes



Women's Shoe Clusters

Cluster 0



Cluster 1



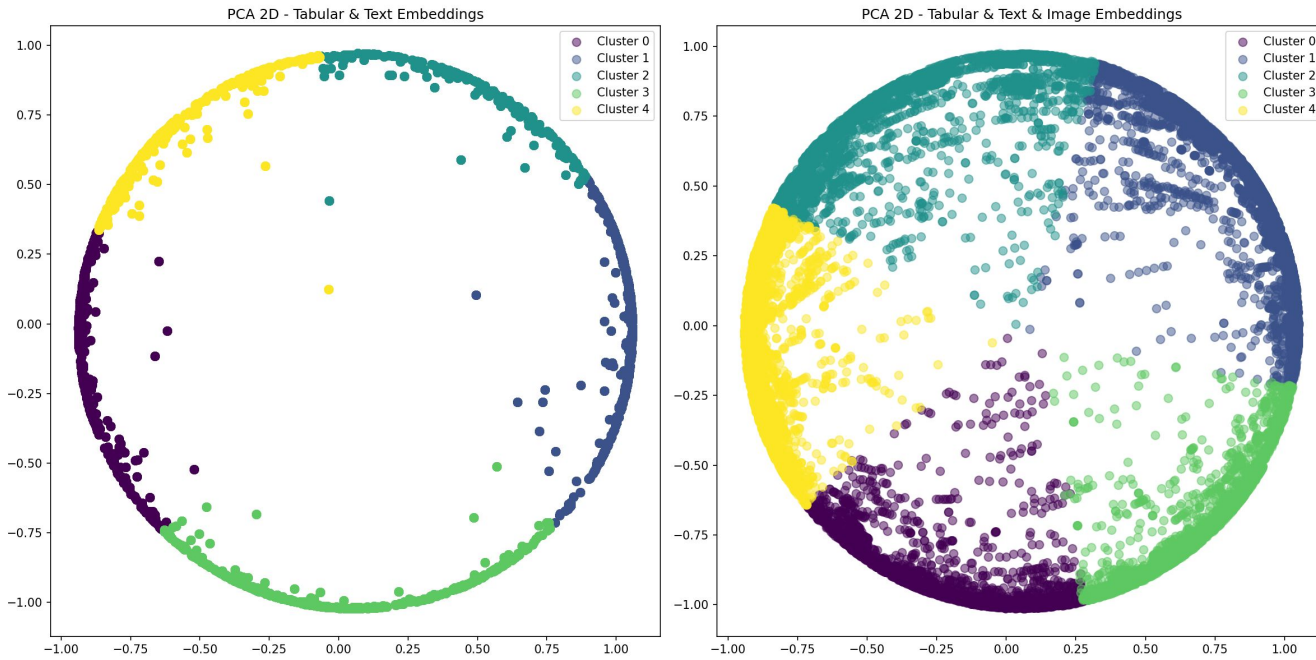
Cluster 2

Cluster 3

Cluster 4

PCA & Product Clustering

KMeans (k=5) | Embedding Space | Men's Shoes



Men's Shoe Clusters

Cluster 0



Cluster 1



Cluster 2



Cluster 3



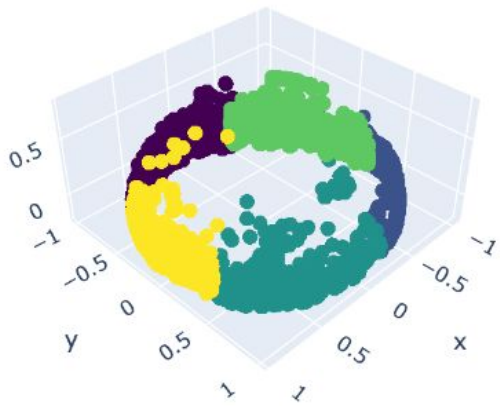
Cluster 4



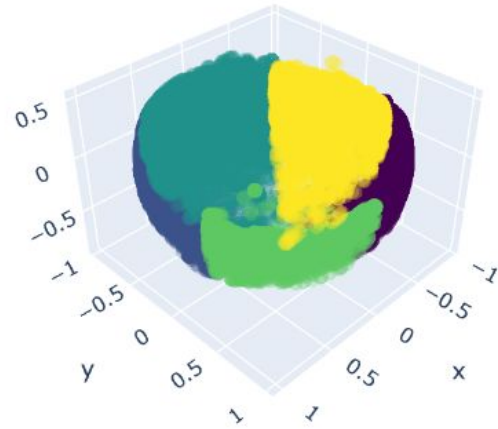
PCA 3D Cluster Visualization

Women's

Tabular & Text Embeddings

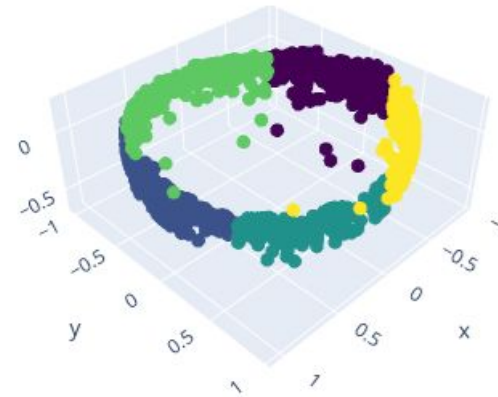


Tabular & Text & Image Embeddings

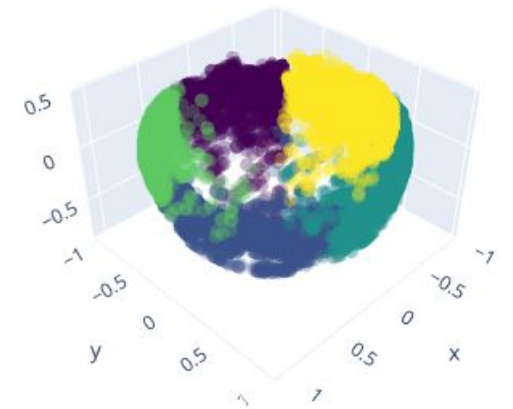


Men's

Tabular & Text Embeddings



Tabular & Text & Image Embeddings

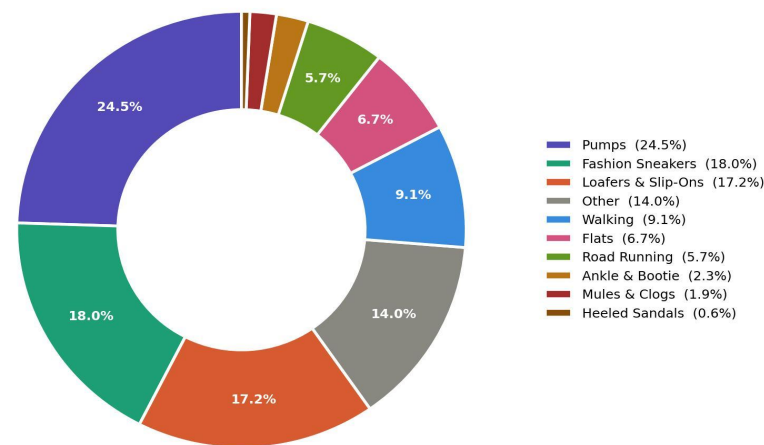


Key Observations

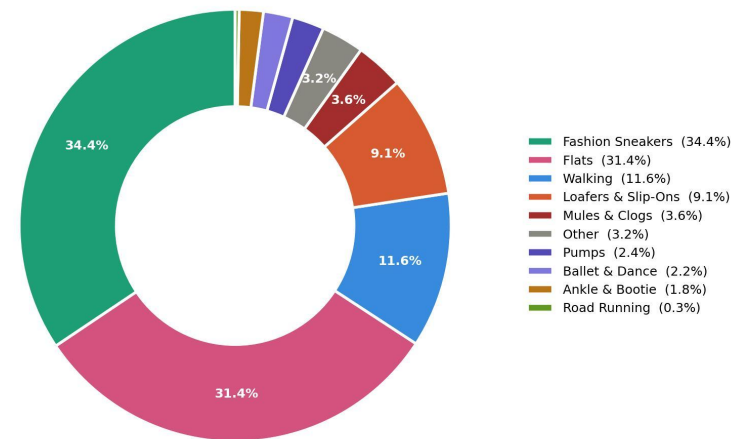
Unlike Victor et al. (2025) where images added little for toy cars, shoes are highly visual ==> color, style, and silhouette matter to buyers. Adding images produces visibly tighter, better-separated clusters, meaning the model understands product differences more accurately. Both women's and men's segments, confirming image modality is essential for fashion categories.

Cluster Distribution for Women Size 8

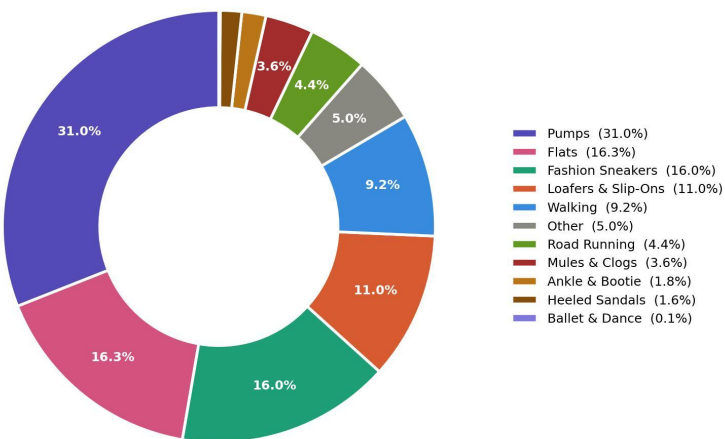
Cluster 0



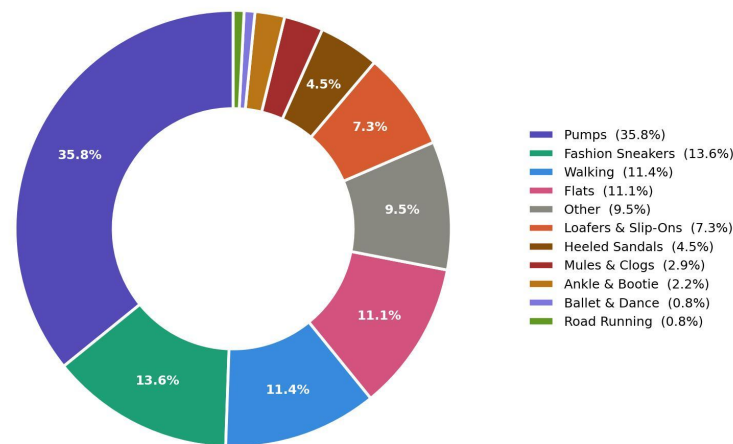
Cluster 1



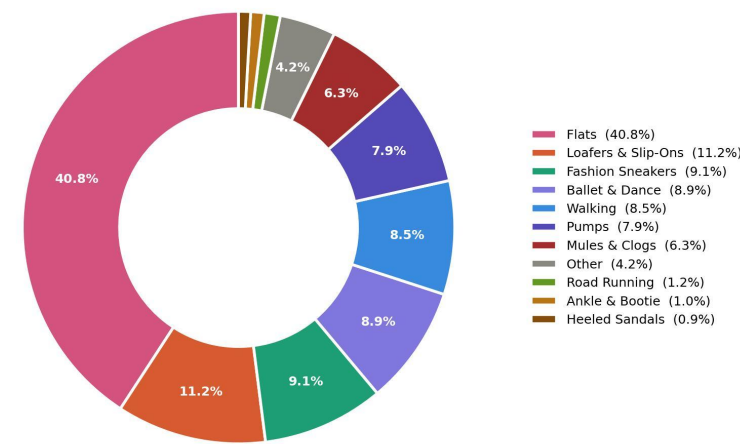
Cluster 2



Cluster 3

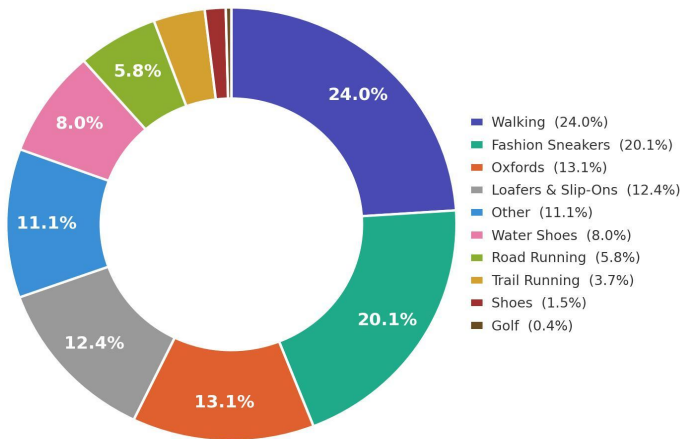


Cluster 4

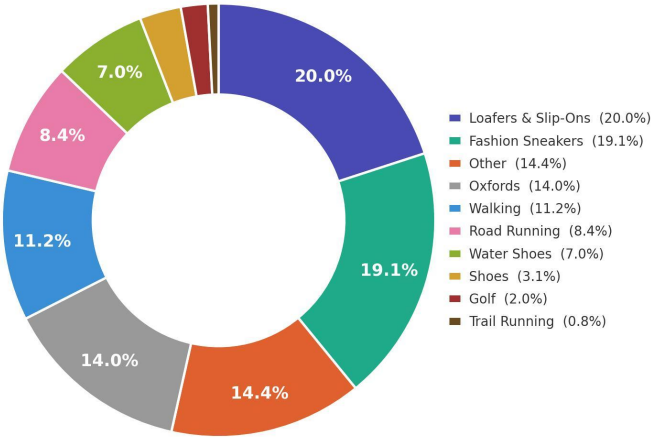


Cluster Distribution for Men Size 8

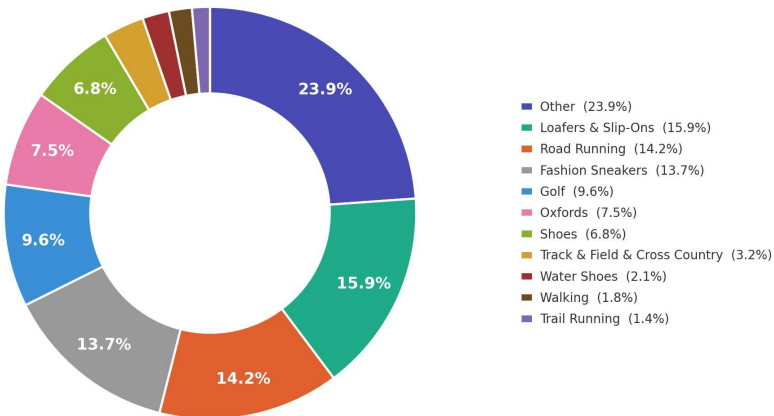
Cluster 0



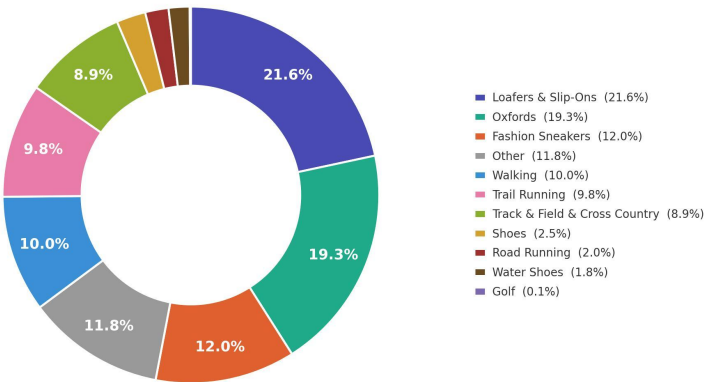
Cluster 1



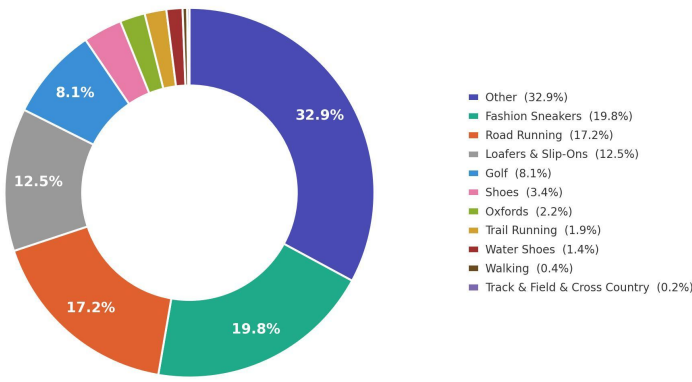
Cluster 2



Cluster 3



Cluster 4



Double Machine Learning (DoubleML)

Causal Inference for Price Elasticity

The Challenge

"If price goes up 1%, how does demand change?"

Naive regression is biased: brand, quality,

reviews all affect BOTH price and demand

We need to isolate the true causal effect

DoubleML Solution

Stage 1: ML predicts sales rank AND price using embeddings as controls. Save residuals.

Stage 2: Regress residuals on each other.

Removes confounding, isolates true effect.

Why AI Embeddings as Controls?

Traditional controls (brand, category) miss subtle product characteristics. Multimodal embeddings capture visual style, text semantics, and market positioning that humans recognize but standard variables cannot encode. Better controls = less omitted variable bias = more accurate elasticity.

Instrumental Variable: Weighted Substitute Price

Distance-weighted average price of 5 nearest neighbors in embedding space, lagged one period. Competitors' prices affect this product's demand (consumers can switch), but don't directly cause this product's price to change. Valid instrument for causal inference.

Women's vs Men's: Predictive Performance — Nuisance Model R²

Side-by-Side DoubleML Results

Women's Shoes

	R2 Q Train	R2 Q Test	R2 P Train	R2 P Test
OLS (Tabular)	0.198218	0.168780	0.240419	0.192626
Boosting (Tabular)	0.592756	0.522182	0.562144	0.404502
OLS (Tabular + PCA)	0.658984	0.575474	0.619038	0.510152
Boosting (Tabular + PCA)	0.902234	0.771662	0.879742	0.690785
OLS (Tabular + Similarities)	0.676718	0.587811	0.628433	0.514226
Boosting (Tabular + Similarities)	0.882863	0.756810	0.842498	0.651803
OLS (Tabular + Embeddings)	0.830711	0.708981	0.789709	0.638545
Boosting (Tabular + Embeddings)	0.918809	0.778197	0.909382	0.681802
Deep Time Independent	0.822300	0.716300	0.775600	0.639000
Deep Time Dependent	0.902100	0.891300	0.779600	0.644800

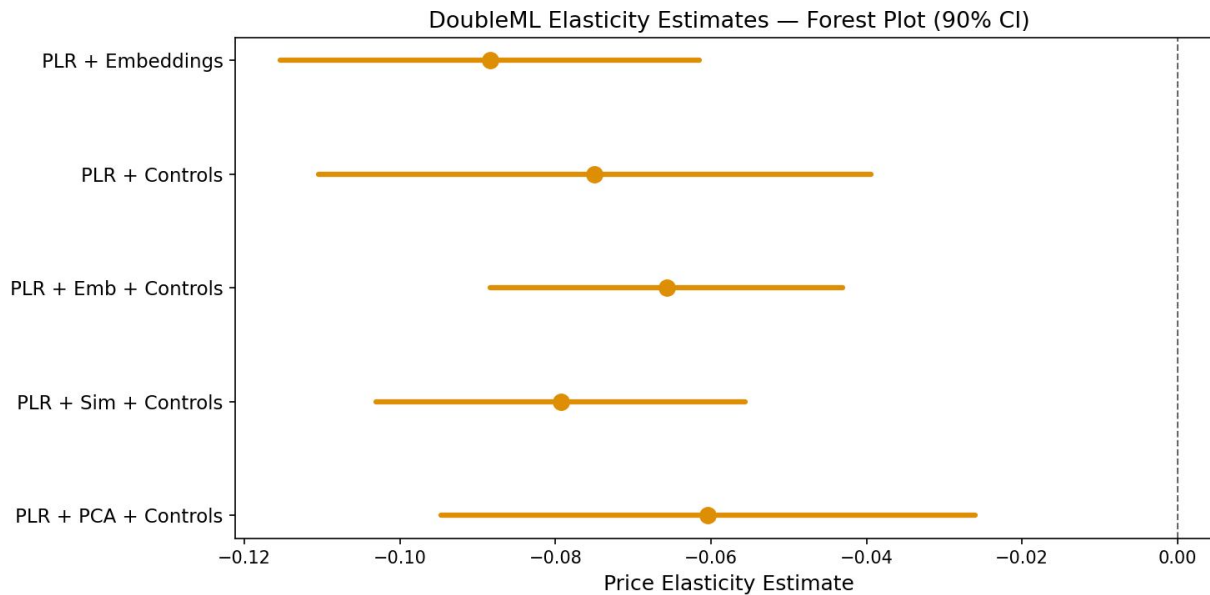
Men's Shoes

	R2 Q Train	R2 Q Test	R2 P Train	R2 P Test
OLS (Tabular)	0.198218	0.168780	0.240419	0.192626
Boosting (Tabular)	0.592756	0.522182	0.562144	0.404502
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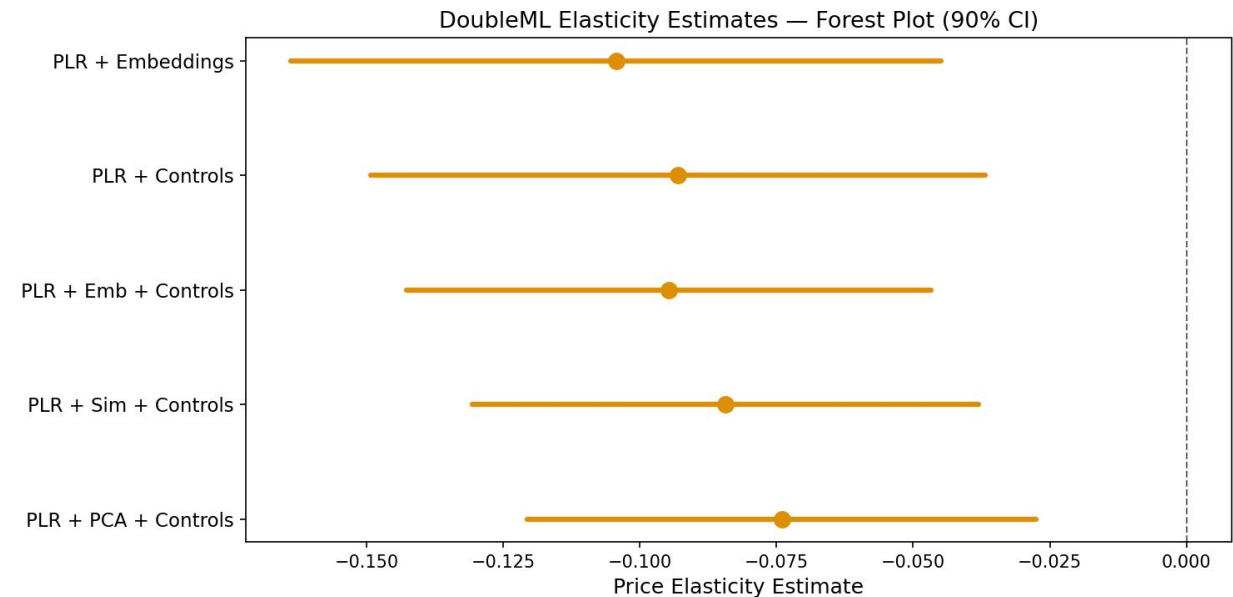
Women's vs Men's: Homogeneous Elasticity Comparison

Side-by-Side DoubleML Results

Women's Shoes (5,147 ASINs)



Men's Shoes (2,061 ASINs)



Key Observations

Both segments show significant negative price elasticity, confirming the approach works across genders

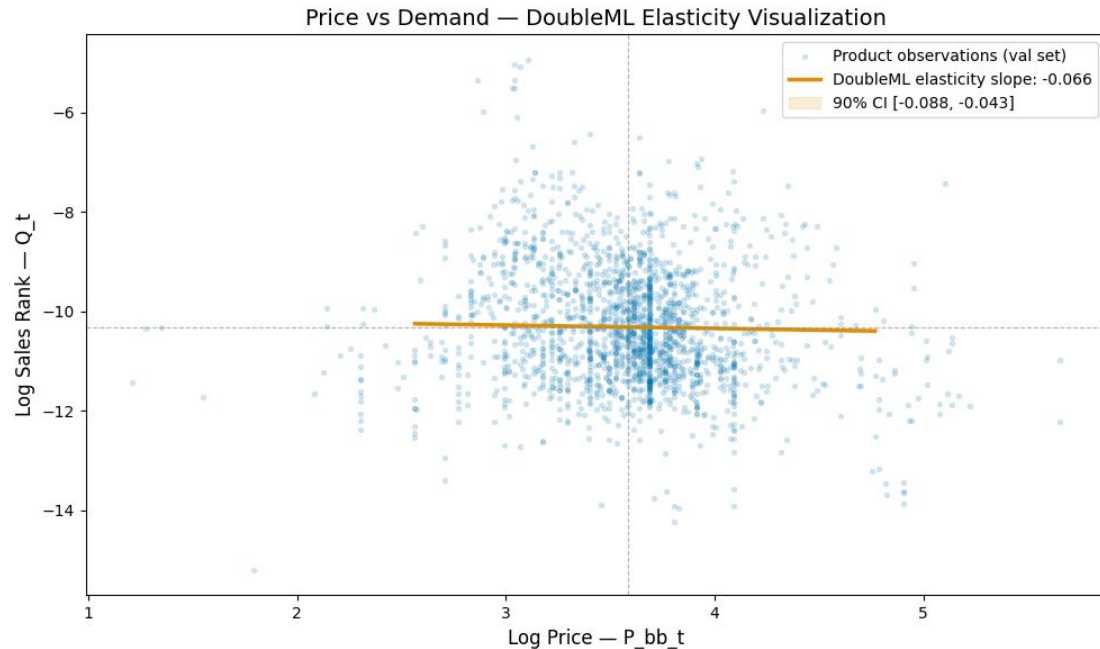
Identical methodology, only the data differs, making this a clean apples-to-apples comparison

Different category structures (Pumps/Heels vs Oxfords/Loafers) lead to different cluster dynamics

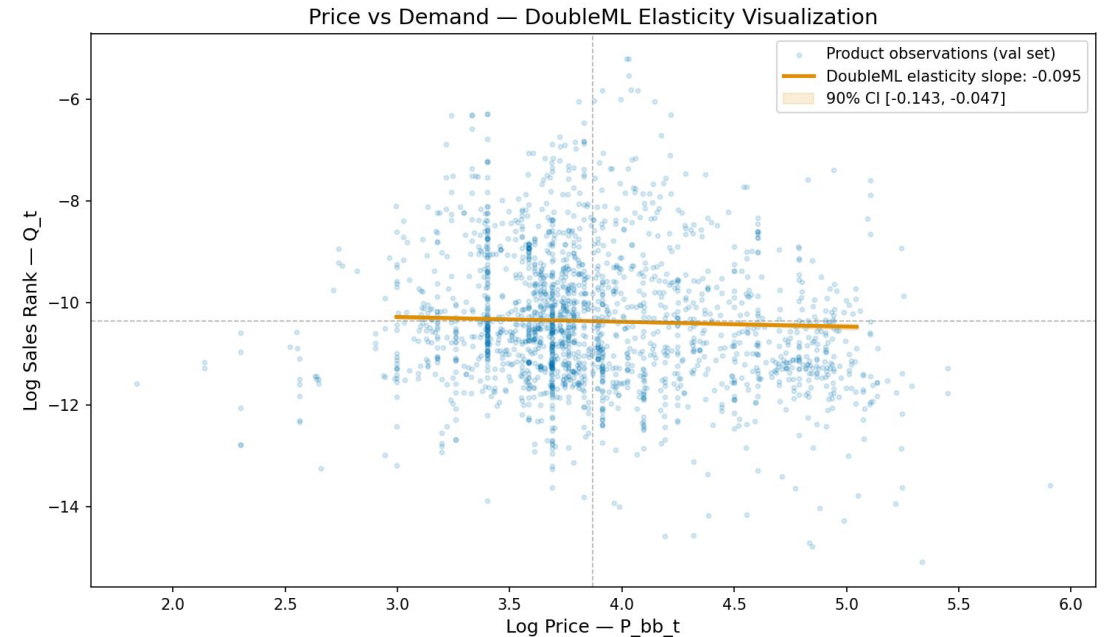
Women's vs Men's: Homogeneous Elasticity Slope

Side-by-Side DoubleML Results

Women's Shoes (5,147 ASINs)



Men's Shoes (2,061 ASINs)



Key Observations

Men's shoes (-0.095) are slightly more price sensitive than women's (-0.066), but both sit in the inelastic range.

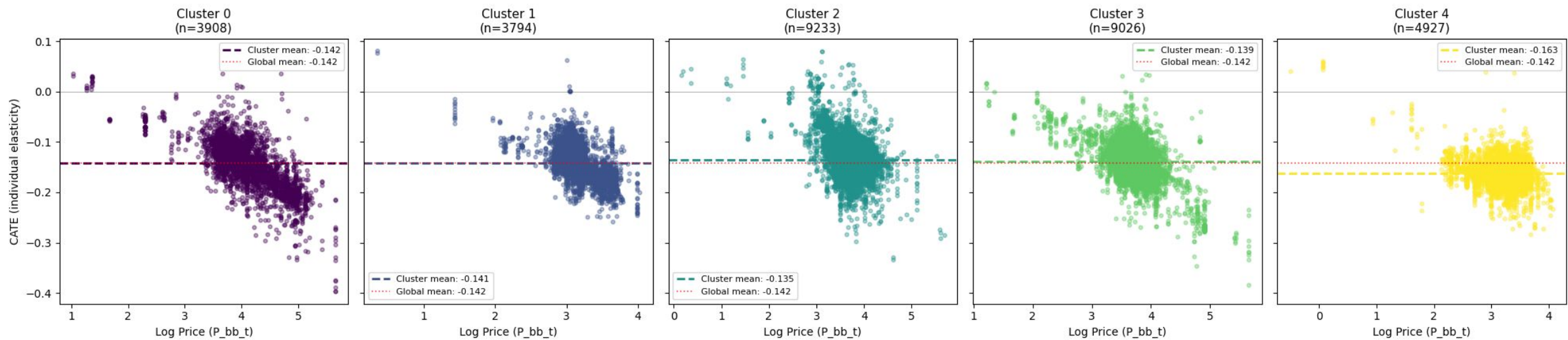
Both segments show statistically significant negative elasticity, confirming DoubleML works consistently across genders.

Identical methodology applied to both segments makes this a clean apples-to-apples causal comparison.

Heterogeneous Effects (CATE)

How Price Sensitivity Varies Across Product Clusters | Women's Shoes

CATE vs Log Price — Separated by Cluster
(red dotted = global mean, dashed = cluster mean)



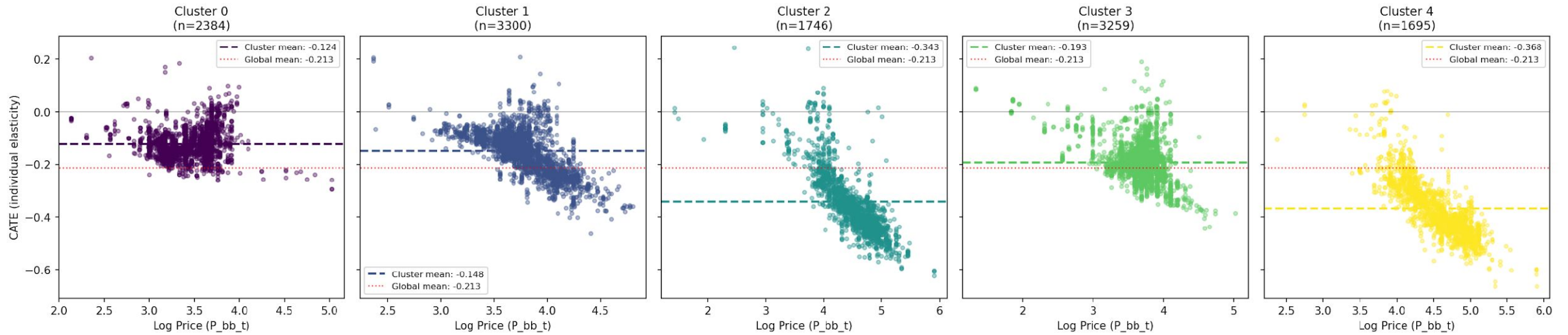
Price Sensitivity by Cluster

Cl. 0	\$57	-0.14	Cl. 1	\$24	-0.14	Cl. 2	\$38	-0.13	Cl. 3	\$40	-0.13	Cl. 4	\$23	-0.16
-------	------	-------	-------	------	-------	-------	------	-------	-------	------	-------	-------	------	-------

Premium shoes (\$87-94) are 3x more price-elastic than budget shoes (\$31)

How Price Sensitivity Varies Across Product Clusters | Men's Shoes

CATE vs Log Price — Separated by Cluster
(red dotted = global mean, dashed = cluster mean)



Price Sensitivity by Cluster

Cl. 0	\$31	-0.12	Cl. 1	\$42	-0.15	Cl. 2	\$87	-0.34	Cl. 3	\$39	-0.19	Cl. 4	\$94	-0.37
Budget walking			Loafers, dress			Premium athletic			Mixed casual			Premium perf.		

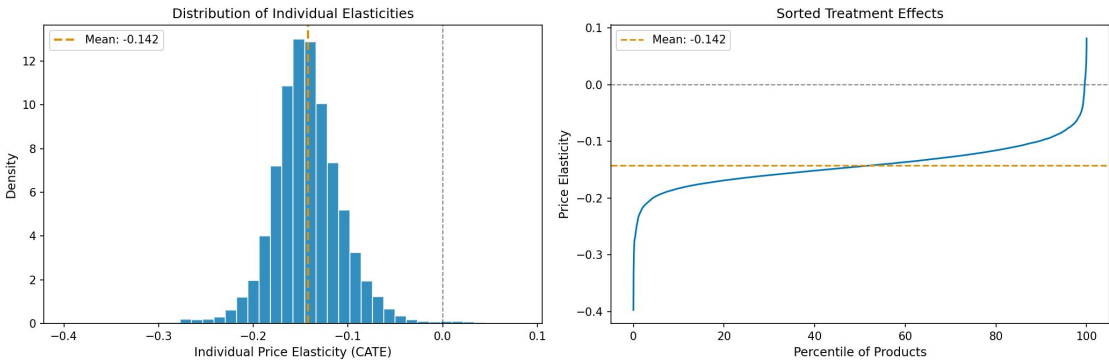
Premium shoes (\$87-94) are 3x more price-elastic than budget shoes (\$31)

CATE Distributions: Women vs Men

Heterogeneous Treatment Effects Across Gender Segments

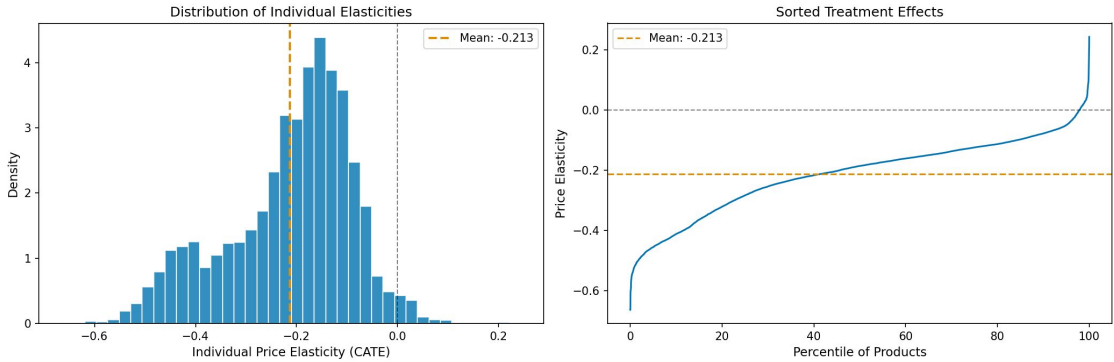
Women's CATE Distribution

Heterogeneous Price Elasticities — CATE Distribution

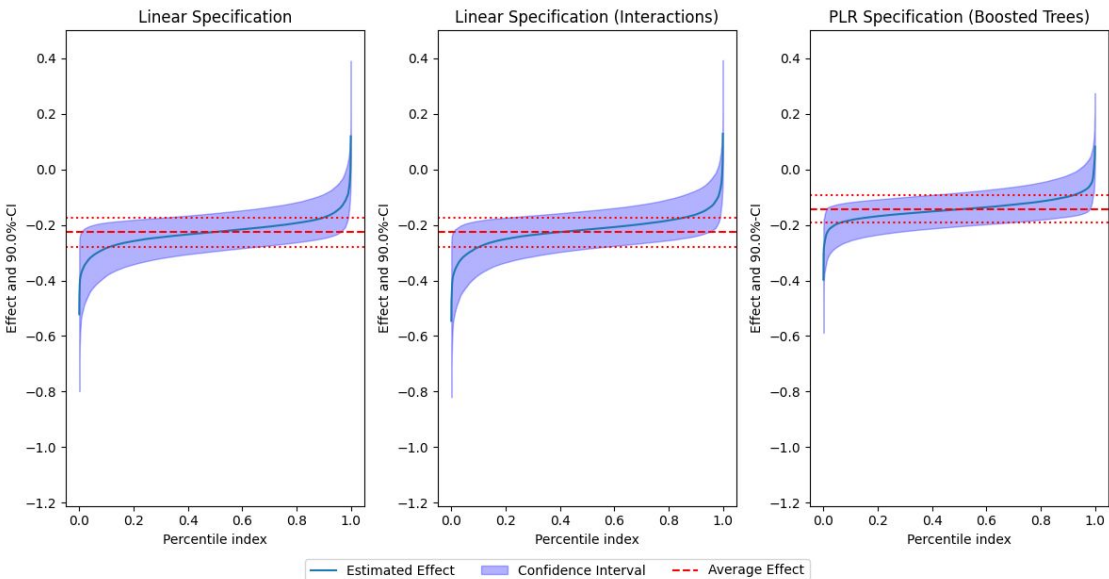


Men's CATE Distribution

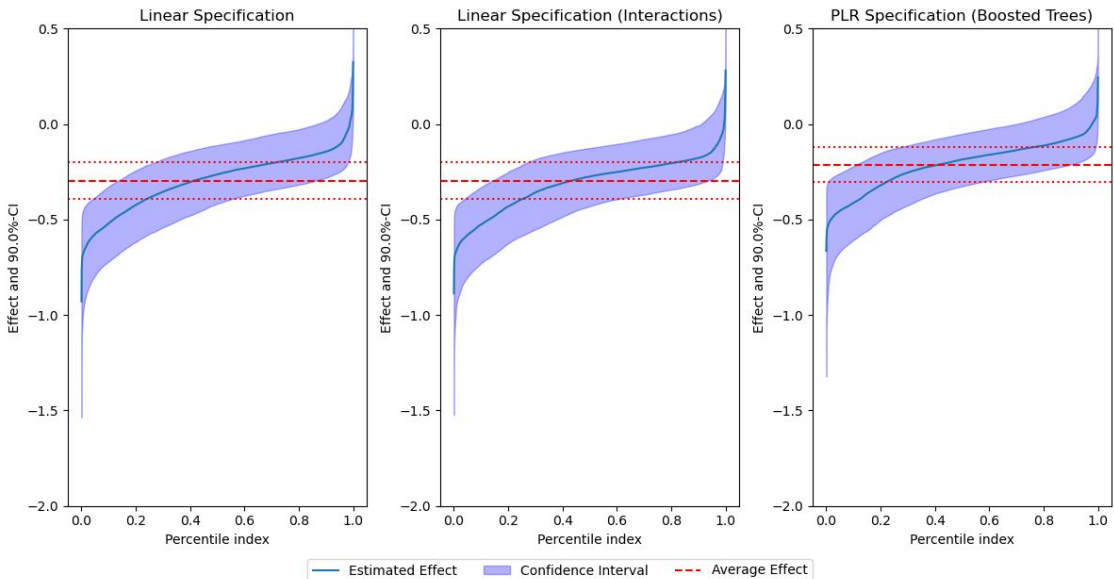
Heterogeneous Price Elasticities — CATE Distribution



Women's Cluster Sorted Effects



Men's Cluster Sorted Effects



Next Steps & Future Directions

Expand to More Sizes

Apply pipeline to sizes 7, 9, 10 to test if elasticity patterns hold

Cross-Gender Statistical Tests

Formally test whether Women vs Men elasticities are significantly different

Temporal Dynamics

Analyze seasonal effects: holiday pricing, back-to-school, Prime Day impacts

Category-Level DoubleML

Separate models per subcategory for finer-grained pricing insights

Dynamic Pricing Application

Build a recommendation system using estimated elasticities for real-time pricing

Additional Data Sources

Incorporate customer review text, Q&A data, and competitor pricing signals

Thank You!

Questions & Discussion

Billy | Ian | Yuhui | Jyoti

Advisor: Dr Guy Arie, Dr Ruidong Ma

Victor et al. (2025) "Adventures in Demand Analysis Using AI"

Northeastern University | April 2026

